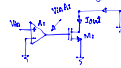
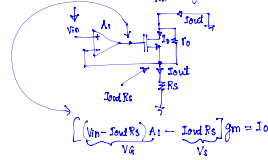


gain boosting:



$$I_{out} = g_m A_1 V_{in}$$

$$g_m = \frac{I_{out}}{V_{in}} = A_1 g_m$$



$$I_{out} = I_0 \frac{r_o}{r_o + R_s}$$

$$I_{out} = [(V_{in} - I_{out} R_s) A_1 - I_{out} R_s] g_m \frac{r_o}{r_o + R_s}$$

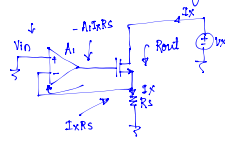
$$I_{out} [1 + (A_1 R_s + R_s) g_m \frac{r_o}{r_o + R_s}] = V_{in} A_1 g_m \frac{r_o}{r_o + R_s}$$

$$\frac{I_{out}}{V_{in}} = \frac{A_1 g_m r_o}{1 + (1 + A_1) R_s g_m \frac{r_o}{r_o + R_s}}$$

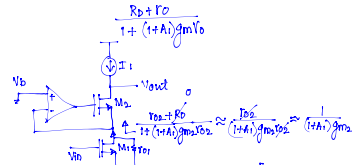
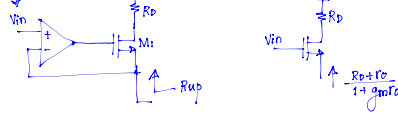
$$= \frac{A_1 g_m r_o}{R_s + r_o + (1 + A_1) R_s g_m r_o}$$

$$= \frac{A_1 g_m r_o}{R_s + r_o + [(1 + A_1) R_s g_m + 1] r_o}$$

$$\approx \frac{A_1 g_m}{R_s + (1 + A_1) g_m r_o R_s}$$



$$R_{out} = R_s + r_o + (1 + A_1) g_m R_s r_o$$



$$I_{out} = g_m V_{in} \times \frac{r_{o1}}{r_{o1} + (1 + A_1) g_{m2} r_{o2}}$$

$$= g_{m1} V_{in} \cdot \frac{r_{o1}}{r_{o1}}$$

$$g_m = \frac{I_{out}}{V_{in}} \approx g_{m1}$$

$$R_{out} = \frac{r_{o1} + r_{o2} + (1 + A_1) g_{m2} r_{o1} r_{o2}}{(1 + A_1) g_{m2} r_{o1} r_{o2}}$$

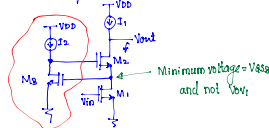
$$\approx \frac{r_{o1} + r_{o2}}{(1 + A_1) g_{m2} r_{o1} r_{o2}}$$

$$Gain = G_m R_{out}$$

$$\approx g_{m1} (1 + A_1) g_{m2} r_{o1} r_{o2}$$

$$\approx (g_m r_o)^3 \text{ if all } g_m \text{ and all } r_o \text{ are equal}$$

Circuit implementation:-

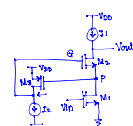


$$Gain = G_m R_{out}$$

$$= g_{m1} (1 + A_1) g_{m2} r_{o1} r_{o2}$$

$$= g_{m1} (1 + g_{m2} r_{o2}) g_{m2} r_{o1} r_{o2}$$

$$V_{out,min} = V_{ds3} + V_{ov2} \text{ --- disadvantage}$$



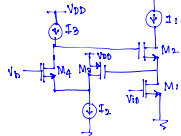
for M_3 to be in saturation

$$V_{ds3} < V_{ds3} - V_{th3}$$

$$V_G \leq V_P - V_{th3}$$

$$V_G + V_{ds2} \leq V_P + |V_{th2}|$$

$$V_{ds2} \leq |V_{th2}|$$



$$A_V = G_m \times R_{out}$$

$$= g_{m1} \times (1 + A_1) g_{m2} r_{o1} r_{o2}$$

$$A_1 = g_{m2} g_{m3} r_{o2} r_{o3}$$